PAPER CODE	EXAMINER	DEPARTMENT	TEL
CPT 107	K.L. Man	Discrete Mathematics and Statistics	1509

2025/26 SEMESTER 1 – Assessment I

BACHELOR DEGREE – Year 2

Discrete Mathematics and Statistics

DEADLINE: 29 October 2025 at 11:59 pm

INSTRUCTIONS TO CANDIDATES

1. **The Assessment should be done individually.**
2. **Total marks available are 100, accounting for 10% of the overall module marks.**
3. **The number in the column on the right indicates the marks for each question.**
4. **Answer all questions.**
5. **Answers should be written in English.**
6. **Relevant and clear steps should be included in your answers.**
7. **Your solutions should be submitted electronically through the Learning Mall via the submission link.**
8. **The naming of Report (in pdf) is as follows: CPT107_StudentID_002.pdf (e.g CPT107_12345678_002.pdf).**
9. **Answers can be handwritten, they should be fully and clearly scanned or photographed for submission as one single PDF document through the Learning Mall via the submission link.**

Notes:

- **To obtain full marks for each question, relevant and clear steps must be included in the answers.**
- **Partial marks may be awarded depending on the degree of completeness and clarity.**

QUESTION 1: Proof Techniques (20 marks)

(a) Prove or disprove that there does exist some integers x and y such that $2x + 6y = \frac{1}{3}$.

(3 marks)

(b) Let a be a positive real number, and the Statement S be: "if a is not rational number, then $\sqrt[6]{a}$ is also not a rational number".

1. Prove or disprove the Statement S.
2. If you think the converse of the Statement S is true, prove it. If not, give a counter-example.

(8 marks)

(c) For x, y, z ∈ ℤ, use proof by contradiction to show the following statement:

if $x^2 + y^2 = z^2$, then x and y are not both odd.

(5 marks)

(d) For all integer $m \geq 1$, use proof by *induction* to show that:

$$m\left(\frac{4m+1}{3}\right)(2m+1) = (1^2 + 2^2 + 3^2 + \ldots + (2m)^2).$$

(4 marks)

QUESTION 2: Set Theory (32 marks)

(a) Let the universal set be $\mathbb{N}$, $\mathrm{A} = \{x \in \mathbb{N} \mid x \text{ is even and } 3 < x < 11\}$, $\mathrm{B} = \{x \in \mathbb{N} \mid x \text{ is odd and } 2 < x < 10\}$, $\mathrm{C} = \{x \in \mathbb{N} \mid x \text{ is odd and } 3 < x < 9\}$ and $\mathrm{D} = \{\mathrm{d} \in \mathbb{N} \mid \mathrm{d}^2 = 7\}$. Find the elements of the following expressions:

1. $\mathrm{B} \cap (\mathrm{A} \cup \mathrm{D})$
2. $\sim((\mathrm{B} - \mathrm{D}) \cap \mathrm{A})$
3. $(\mathrm{C} \cup \mathrm{D}) \cap (\mathrm{B} \,\Delta\, \mathrm{A})$
4. $\big(((\mathrm{D} \cup \mathrm{B}) \cap (\mathrm{A} \,\Delta\, \mathrm{B})\big) \cup (\mathrm{B} - (\mathrm{A} - \mathrm{D}))) \cap \mathrm{C}$

(8 marks)

(b) Let X and Y be sets, then $X \subseteq Y$ if and only if $\sim X \subseteq \sim Y$. If you think that it is true, prove it. Otherwise, give a counterexample to show that it is false.

(6 marks)

(c) Prove that the following sets are equal:

1. $\mathrm{A} \times (\sim(\sim\mathrm{B} \cap \sim\mathrm{C})) = (\mathrm{A} \times \mathrm{B}) \cup (\mathrm{A} \times \mathrm{C})$
2. $\mathrm{A} \times (\mathrm{B} - \mathrm{C}) = (\mathrm{A} \times \mathrm{B}) - (\mathrm{A} \times \mathrm{C})$

(10 marks)

(d) Let X and Y be sets, and $\mathrm{X} \subseteq \mathrm{Y}$ if and only if $(\mathrm{Y} - \mathrm{X}) \cup \mathrm{X} = \mathrm{Y}$. If you think that it is true, prove it. Otherwise, give a counterexample to show that it is false.

(8 marks)

QUESTION 3: Relations (48 marks)

(a) Let $A = \{a \in \mathbb{Z} \mid a \neq 0\}$ and R be the relation on the set A defined by $R = \{(a,b) \mid a + b = 2c\}$, where a,b,c $\in A$. Prove or disprove that R over A is an equivalence notion.

(6 marks)

(b) Let $A = \{a, b, c, d\}$. What is the transitive closure of the relation $\{(a,b), (b,c), (c,d), (a,c), (a,d), (b,d)\}$ on A? Express your answer using ordered pairs.

(4 marks)

(c) If A and B are both transitive relations, then their intersection is also transitive. If you think that it is true, prove it. If not, give a counter-example.

(8 marks)

(d) Let R and S be two partial orders on a set X, and T is a relation on X such that aTb (i.e. $a,b \in X$) if and only if both aRb and aSb hold. Is T also a partial order on X? Justify your answer with proofs and/or counterexamples

(7 marks)

(e) Given the partial ordering on the following set A = {a, b, c, d, e} as {(a,a), (b,b), (c,c), (d,d), (e,e), (a,b), (a,c), (a,d), (a,e), (d,e)}, draw the corresponding Hasse diagram.

(3 marks)

(f) Let A be the points in the plane (A = R × R). We say two points are equivalent if they are equal distance from the origin. So (x, y) ~ (w, z) if $x^2+y^2 = w^2+z^2$. Show this is an equivalence relation on A and find the equivalent classes.

(10 marks)

(g) For all a, b $\in$ Z+, we define a◎b if a/b $\in$ Z+. Prove or disprove ◎ is a partial order and/or a total order on Z+.

(10 marks)

END OF ASSESSMENT I PAPER